

Metropolitan University, Sylhet
Department of Computer Science and Engineering
Spring Term Final Examination - 2023
Program: B.Sc. in CSE; Batch: 58th
Course: GEd 114: English Language I

Time: 2 hours

Marks: 40

Figures in the right-hand margin indicate full marks.

Read the passage and answer questions 1 and 2:

Many ignore trends in their innovation strategies or adopt a wait-and-see approach and let competitors take the lead. At a minimum, such responses mean missed profit opportunities. At the extreme, they can jeopardize a company by ceding to rivals the opportunity to transform the industry. The purpose of this article is twofold: to encourage managers to think about the opportunities of innovation that trends bring along; and to provide some advice on how to make market research and product development personnel more adept at analyzing and exploiting trends.

One strategy, known as 'infuse and augment', is to design a product or service that retains most of the attributes and functions of existing products in the category but adds others that address the needs and desires unleashed by a major trend. A case in point is the Poppy range of handbags, which the firm Coach created in response to the economic downturn of 2008. The Coach brand had been a symbol of luxury for nearly 70 years, so lowering prices could risk cheapening the brand's image. Therefore, Coach launched the lower-priced Poppy sub-brand and avoided across-the-board price cut.

A more radical strategy is 'combine and transcend'. This entails combining the existing product with attributes that address changes arising from a trend in order to create a novel experience. In 2006, Nike teamed up with technology company Apple to launch Nike+, a digital sports kit comprising a sensor that attaches to the running shoe and a wireless receiver that connects to the user's iPod. The Nike+ sports kit and web interface moved the company from a focus on athletic apparel to a new plane of engagement with its customers.

A third approach, known as 'counteract and reaffirm', involves developing products or services that counteract the negative attributes traditionally associated with such products or services. For example, ME2, a video game created by Canada's iToys, incorporated the traditional physical component of children's play: it contained a pedometer, which tracked and awarded points for physical activity (walking, running, biking, skateboarding, climbing stairs). The child could use the points to enhance various virtual skills needed for the video game. Thus, the video game encouraged physical activity.

1. Answer the questions:

1×4=4

- a) What did the brand Coach do in response to the economic downturn of 2008?
- b) What does "combine and transcend" strategy involve in creating new experience?
- c) How can ME2 players use the points gained from doing physical activities?
- d) Which of the three strategies involves shedding the negative image associated with product types?

2. Summarize the above passage in your own words.

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3. Correct the following sentences:

0.5×10=5

- a) The police was on the alert.
- b) Either the team members or their leader are responsible for this incident.
- c) Only one third of the students was present in the class.
- d) Fifteen kilometers are too long for me to walk.
- e) I do a little bit of cook at weekends.
- f) The teacher caught him chat on his phone in the class.
- g) She goes to jog every morning.
- h) I saw her to talk to your father.
- i) If you get 'A+', I'll consider to buy you a bike.
- j) He apologized for make the mistake.

4. Fill in the blanks with words from the box:

0.5×10=5

since, once, so that, as if, now that, whatever, provided that, whereas, unless, though

- a) He speaks so boastfully ___ he owned the place.
- b) ___ the exams are over, we can finally relax and enjoy our vacation.
- c) She studied diligently ___ she could pass the difficult exam.
- d) ___ he was exhausted, he went on working hard.
- e) ___ you submit the required documents, your application will not be processed.
- f) You work only 30 hours a week, ___ he works over 65 hours.
- g) You can use the conference room at weekends, ___ you book it in advance.
- h) ___ happens, we will face the challenges together as a team.
- i) ___ the rain stops, we can go for a walk in the park.
- j) ___ you were busy, I left without asking you.

5. Make ONE sentence with each of the following phrasal verbs:

0.5×10=5

(a) back off, (b) bear with, (c) carry on, (d) come across, (e) figure out, (f) go about, (g) let down, (h) make out, (i) pull off, (j) set up

6. Answer any ONE of the following:

07

- a) Suppose you stayed at a friend's house while you were on holiday. Write a letter thanking him/her and his/her family for their hospitality.
- b) Suppose IT Hax Ltd. has published a job advertisement in *The Prothom Alo*. They want to recruit a few junior software engineers. Now write a job application for this position.

7. Briefly discuss any ONE of the following topics. Write at least 220 words.

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- a) YouTube as a Learning Tool
- b) Importance of Eating Healthy Food

Metropolitan University, Sylhet
 Department of Computer Science and Engineering
 Spring Semester Final Examination – 2023
 Program: B.Sc. in CSE; Batch – 58
 Course: MAT 112: Differential and Integral Calculus

Time: 2 Hours

Marks: 40

All parts of each questions must be answered sequentially. Figures in the right margin indicate full marks.

Group – A
 (Answer any two questions)

1. (a) Differentiate $\tan^{-1} \frac{2x}{1-x^2}$ with respect to $\sin^{-1} \frac{2x}{1+x^2}$. 5
- (b) If $\log y = \tan^{-1} x$, prove that 5
- $$(1+x^2)y_{n+2} + (2nx + 2x - 1)y_{n+1} + n(n+1)y_n = 0$$
2. (a) If $f(x) = \begin{cases} 1; & x < 0 \\ 1 + \sin x; & 0 \leq x < \frac{\pi}{2} \\ 2 + \left(x - \frac{\pi}{2}\right)^2; & x \geq \frac{\pi}{2} \end{cases}$, discuss the differentiability at $x = \frac{\pi}{2}$. 5
- (b) If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, then show that 5
- $$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x+y+z)^2}.$$
3. (a) Verify the application of Rolle's theorem for the function $f(x) = x^2$ in the interval $(-1, 1)$. 5
- (b) Verify mean value theorem for the function $f(x) = x^3 + 3x^2 - 2x$ in the interval $[-2, 2]$. 5

Group – B
 (Answer any two questions)

4. Evaluate the integral (Any two) 10
- (a) $\int \frac{\cos x}{a+b^2 \sin^2 x} dx$ (b) $\int \frac{dx}{\sqrt{x^2-7x+12}}$ (c) $\int \frac{2x+7}{2x^2-2x+1} dx$
5. (a) Define Beta function and Gamma function. Prove that $\Gamma(n+1) = n\Gamma n$. 5
- (b) Prove that $\int_0^{\frac{\pi}{2}} \sin^5 \theta \cos^4 \theta d\theta = \frac{8}{315}$ 5
6. (a) Show that $\int_0^{\frac{\pi}{2}} \frac{d\theta}{1+\tan \theta} = \frac{\pi}{4}$ 5
- (b) Find the area enclosed by the parabola $y^2 = 8x$ and the straight line $4x - y - 4 = 0$. 5

Metropolitan University, Sylhet
 Department of Computer Science and Engineering
 Spring Term Final Examination – 2023
 Program: B.Sc. in CSE; Batch: 58
 Course: CSE 123: Discrete Mathematics

Time: 2 Hours

Marks: 40

Answer any **four** questions. All parts of each question must be answered sequentially.

Figures in the right margin indicate full marks.

- ① a) Determine a name of this graph, number of vertices, edges, and find the in-degree and out-degree of each vertex for the given graph, presented in Figure 1. 4

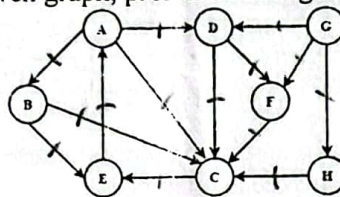


Figure 1. Graph

- b) Use adjacency matrix and adjacency list to describe the graph in Figure 1. 2 + 2
- c) Traverse the following graph presented in Figure 2 using the BFS algorithm, while starting node is A. 2

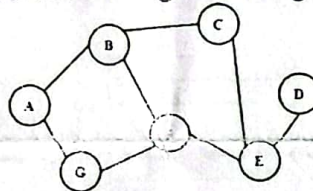


Figure 2. Graph

- a) Draw a minimum spanning tree from the following graph (Figure 3) based on Kruskal's algorithm. 4

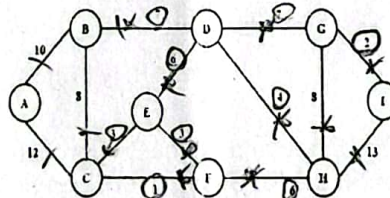


Figure 3. Graph

- b) Find the shortest path from the vertex "A" to vertex "F" using the Dijkstra's shortest path algorithm in the following figure (Figure 4). 4

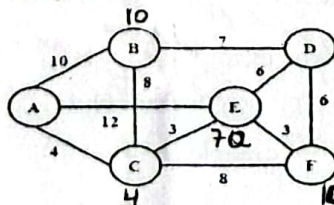


Figure 4. Graph

- c) Write short notes on spanning tree and minimum spanning tree. 2

- ③ a) Consider the following tree in Figure 5, which is a binary tree. Write down the name of the nodes in the order they are reached if you perform in-order/infix, pre-order/prefix and post-order/postfix traversal operations on this tree, where starting node is A. 4

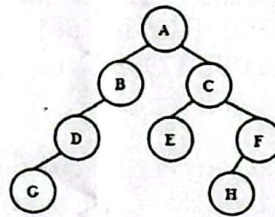


Figure 5. Tree

Pre-order: A B D G E C F H

In-order: G D B A E C H F

Post-order: G D B E H F C A

- b) Draw a Huffman Tree using the following frequency table – 2 + 2

A	E	I	O	U
22	35	23	12	29

Table 1. Frequency Table

Calculate the total number of bits required to represent these characters.

- c) What are key differences between Tree and Graph? 2

- ④ a) List the types of fundamental logic gates and explain any two of them with truth tables and circuit diagrams. 1 + 3

- b) Show that the Distributive Law $a(b + c) \equiv ab + ac$ is valid using a truth table. 4

- c) Find the output of this circuit - 2

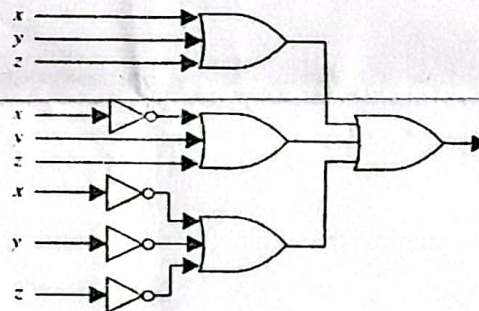


Figure 6. Logic Circuit

5. a) What is the key objective of the divide and conquer approach? Write step-by-step processes to perform Merge sort operation in the following data elements: 37, 26, 41, 3, 8, 81, 16. 1 + 3

- b) Draw a binary search tree using the following data elements: 55, 51, 81, 10, 55, 12, 20, 59, 102, 22, 30. Do the following operations on this binary search tree: insert a data: 99 and delete root node. 2 + 2

- c) Write short notes on SOP and POS. 2

6. a) Explain a Recursion function with an example. 4

- b) Write short notes on the three asymptotic notations O , Ω and Θ . 4

- c) What are key differences between Time and Space complexities? 2

O b c $b+c$ $a(b+c)$

a b c cb bc a

66
70
46
326
58
276

Metropolitan University
Department of Computer Science and Engineering
Spring Term Final Examination – 2023

Program: CSE: Batch: 58
Course: PHY 111: Physics I
Set: B₂

Time: 2 Hours

Marks: 40

[Answer any four of the following questions]

1. (a) What are the characteristics of a simple harmonic motion? Draw the energy diagram of the simple harmonic motion. 2+2
- (b) A block of mass is attached to a spring of spring constant k and is in simple harmonic motion. The amplitude of the motion is A . Explain 3+3
- i. Why the potential energy is zero at $x = 0$?
- ii. Why the kinetic energy is the maximum at $x = 0$?
2. (a) The displacement equation of a wave is given below:
- $y = (6\text{mm}) \sin(3\pi x - 2t)$
- (i) Calculate the wave number, time period, and velocity of the wave at $x = 0$ and $t = 0$ 3+3
- (ii) Draw the graph of the velocity.
- (b) Find the solution of the equation of motion of a simple harmonic oscillator. 4
3. (a) Why do lakes freeze on the top rather than at the bottom? 2
- (b) What are (a) constant-volume (b) isobaric, and (c) isothermal processes? Show these processes using a PV diagram. 4
- (c) A 10 kg block extends a spring 10 cm from its unstretched position. Now another 5 kg object is hanging from the same spring. Find the displacement and the period of the oscillation. 4
4. (a) Define internal energy E_{int} of a thermodynamic system. What happens to the first law during an isothermal process? 2+2
- (b) Let 1.00 kg of ice be converted to steam by melting at standard pressure (100 kPa). The volume changes from an initial value of 2 m^3 as a solid ice to 2.50 m^3 as liquid water. For this process, find (a) the work done on the system, (b) the heat added to the system, and (c) the change in the internal energy of the system. (The heat of transformation for ice is 336000 J/kg) 6

5 (a) State the Zeroth law of thermodynamics.

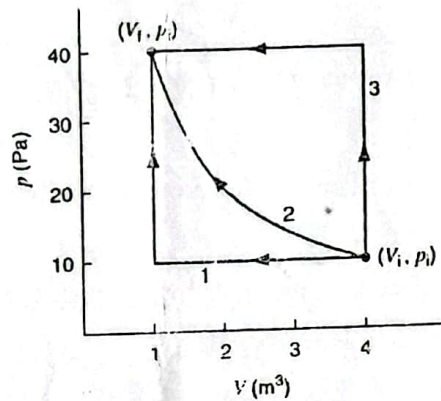
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(b) How can you define temperature using the concept of the Zeroth law of thermodynamics?

2

(c) A sample of gas consisting of 6 mol is compressed from a volume of 7.0 m^3 to 3.0 m^3 while its pressure increases from 20 to 70 Pa. Compare the work done along the paths 2 and 3 as shown in the figure below.

6



6 (a) Draw the PV diagram for the Carnot's cycle and explain the steps using the PV diagram.

6

(b) A Carnot engine takes 2000 J of heat from a reservoir at 823 K, does some work, and discards some heat to a reservoir at 323 K.

4

i. How much work does it do?

ii. Calculate the efficiency of the engine.

Metropolitan University, Sylhet
Department of Computer Science and Engineering
Spring Term Final Examination – 2023
Program: B.Sc. in CSE; Batch: 58(A-I)
Course: GED 201 : Bangladesh Studies

Time: 2 Hours

Marks: 40

Answer any **four** questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

1. (a) Write the historical significance of "The Battle of Plassey of 1757." 5
(b) Outline the European intervention in the Indian subcontinent. 5
2. (a) Define Industry and Sectors of Economy, also show the difference between GDP and GNP. 6
(b) "Leather and IT industries are the next bright stars in the Bangladesh Economy." comment on the statement. 4
3. (a) Define and distinguish equity and equality, write in your own words- requirements for women empowerment in the contemporary time? 6
(b) Is women's representation in politics satisfactory? Justify in the context of Bangladesh. 4
4. (a) Define "Blue Economy" and write what are the promising sectors of the blue economy in the context of Bangladesh with short description of each. 6
(b) "Sustainable and eco-friendly use of sea resources should be the key motivation behind economic development of Bangladesh"- develop an argument for the statement. 4
5. (a) Can the current demographic dividend of Bangladesh turn into a resource curse? Shade light on the facts of current demographic dividend of Bangladesh on the basis of the statement. 6
(b) In a particular country, there were 2,500 live births in a year, and the area resident population was 200,000. What was the birth rate in that country during that year? 4
6. (a) Discuss the pillars of Smart Bangladesh. 5
(b) How do you want to see Bangladesh in 2041? 5

Metropolitan University, Sylhet
Department of Computer Science and Engineering
Spring Term Final Examination – 2023
Program: B.Sc. in CSE; Batch: 58 (A-I)
Course: GED 202 : History of Emergence of Bangladesh

Time: 2 Hours

Marks: 40

Answer any **four** questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

- ① (a) "1952 Language movement is the first spark of Bengali nationalism"- justify the statement with proper arguments. 6
- (b) Who gave the title "Bangabandhu" to Sheikh Mujibur Rahman? Write the historic context of the 1969 mass upsurge. 4
- ② (a) What are the six points of the "Six point Programme" why it was called "*Banglair Muktiur Sanad*"? 5
- (b) Was the disparity in economy and basic rights of West and East Pakistan the ultimate cause of two Pakistan's separation? 5
- ③ (a) What was the outcome of the 1970's Pakistan General election? And what impact did it hold in the 1971 liberation war? 5
- (b) Was the 1971 massacre a Genocide? Build your argument? 5
4. (a) Who were the cabinet members of the "Mujibnagar exile government" and write about their roles as the war time government. 5
- (b) Discuss the war strategy of Mukti Bahini adopted in the 1971 Liberation war? 5
5. (a) Describe the final days of the Pakistan Army after 4th of November 1971 to 16th December 1971. 6
- (b) Write a short note on "Surrender at Dhaka" 4
- ⑥ (a) Describe the four fundamental principles of 1972's constitution. 4
- (b) Write the historical significance of the 7th March speech of Bangabandhu in 1971. 6